Akoizo Original Practice - 2027 Rotation Form 1

Division B | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) One mm over one km^2 equals…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. 1 m^3

B. 1 million m^3

C. 1000 liters

D. 1000 m^3

Answer: _____

2. (2 points) Snow depth equals liquid-water depth automatically?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. Only in spring

B. No

C. Always

D. Only in mountains

Answer: _____

3. (2 points) Two second-order streams create…

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Second order

B. Fourth order

C. First order

D. Third order

Answer: _____

4. (2 points) Dissolved load consists of…

Unit: Sediment transport and channel forms.
Worked-case data: A river has Q=20 m^3/s. Compare velocity for areas 10 and 20 m^2.

A. Only sand

B. Ions and molecules in solution

C. Only gravel

D. Only floating wood

Answer: _____

5. (2 points) An oxbow commonly forms through…

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. Only glacier freezing

B. Only coral growth

C. Only groundwater pumping

D. Meander cutoff

Answer: _____

6. (2 points) Hydrograph area represents…

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Only basin area

B. Volume

C. Only peak stage

D. Only velocity

Answer: _____

7. (2 points) High porosity always means high permeability?

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Always

B. Only for clay

C. Only below water table

D. No

Answer: _____

8. (2 points) Every lake mixes twice per year?

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only if freshwater

B. No

C. Always

D. Only if deep

Answer: _____

9. (2 points) Impervious connectivity can affect…

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Only mineral hardness

B. Only gene frequency

C. No water process

D. Runoff timing

Answer: _____

10. (2 points) A contour connects equal…

Unit: Maps, monitoring, and integrated freshwater investigations.
Worked-case data: Two wells have heads 110 and 106 m, separated by 200 m. Find gradient magnitude.

A. Chemical concentration always

B. Elevation

C. Discharge

D. Rainfall always

Answer: _____

11. (2 points) A freshwater catchment receives 110 mm precipitation, loses 40 mm to evapotranspiration and exports 30 mm runoff. Find storage change.

Answer: _____
_____
_____

12. (2 points) A rectangular stream is 5 m wide and 2 m deep with mean velocity 0.5 m/s. Calculate discharge.

Answer: _____
_____
_____

13. (2 points) A river channel is 13 km long between points 9 km apart along the valley. Find sinuosity.

Answer: _____
_____
_____

14. (2 points) A water table drops 2 m over 100 m horizontal distance. Find hydraulic gradient.

Answer: _____
_____
_____

15. (2 points) Use Q=K A i with K=0.001 m/s, A=21 m^2 and i=0.02. Find saturated Darcy discharge.

Answer: _____
_____
_____

16. (2 points) A lake stores 1000 m^3 and exports 100 m^3/day in a steady well-mixed teaching model. Estimate residence time as storage/outflow.

Answer: _____
_____
_____

17. (2 points) An urbanized watershed has more impervious area. Predict one hydrograph change under a comparable storm and explain.

Answer: _____
_____
_____

18. (2 points) A stream bends through its floodplain. Locate likely erosion and deposition relative to the bend.

Answer: _____
_____
_____

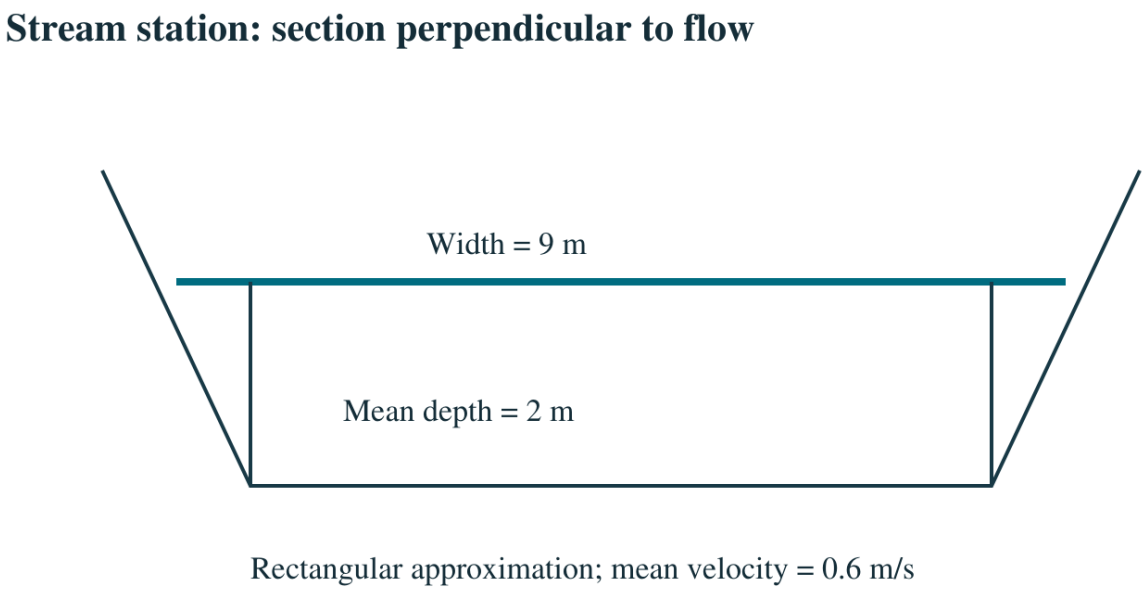
19. (2 points) Two first-order channels join, and no higher-order channel enters. What Strahler order results, and why?

Answer: _____
_____
_____

20. (2 points) Groundwater pumping increases near a connected river. Explain a possible effect on river baseflow.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Estimate wetted cross-sectional area using the indicated rectangular approximation.



F1. Stream cross section and measured velocity (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Calculate stream discharge from this section.

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

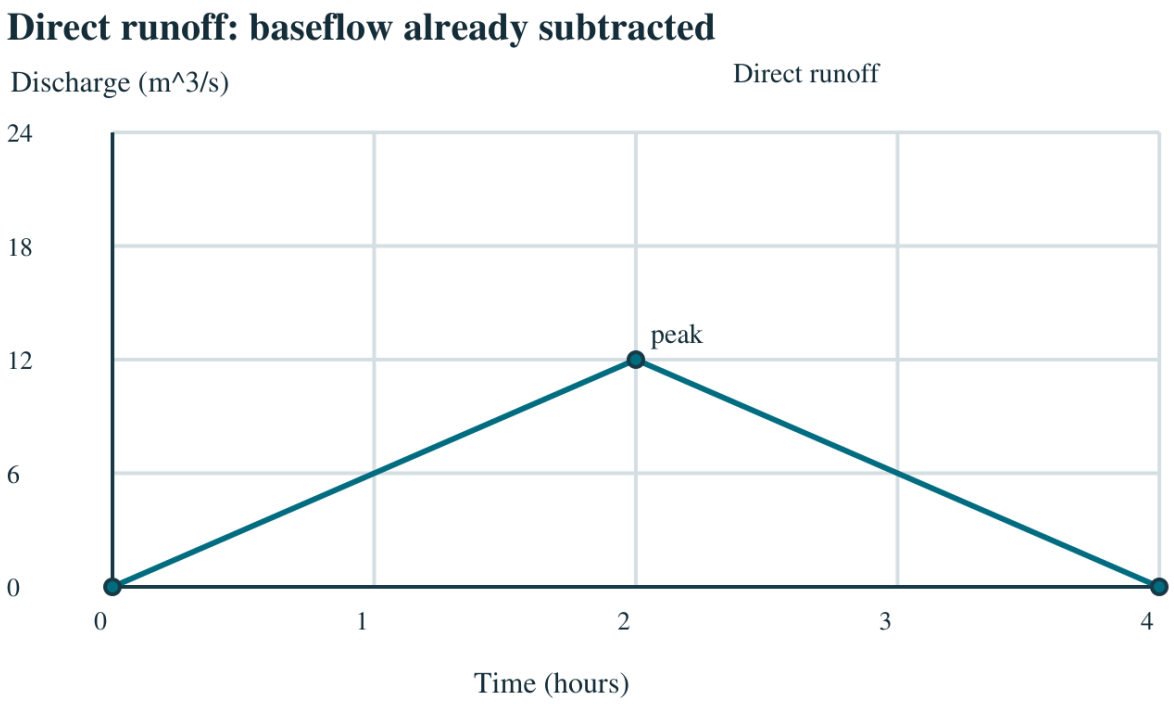
Answer: _____
_____
_____

23. (2 points) Use Figure F1. The water rises while width and mean velocity remain fixed. Explain the predicted discharge change.

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Read the time of peak direct runoff.



F2. Storm hydrograph (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Find the event runoff volume. Convert hours to seconds.

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

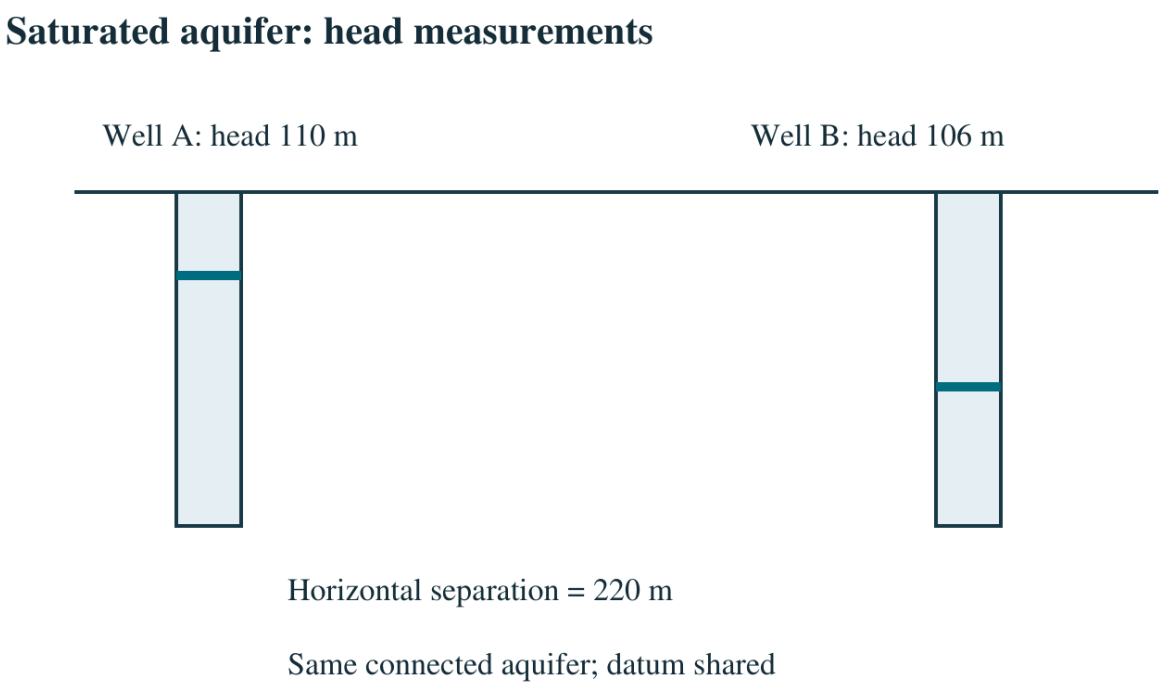
Answer: _____
_____
_____

26. (2 points) Use Figure F2. A wetland spreads the same runoff volume over more time. Describe how its hydrograph could differ.

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Calculate hydraulic-gradient magnitude.



F3. Groundwater heads in two wells (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Predict the horizontal flow direction and identify the controlling measurement.

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

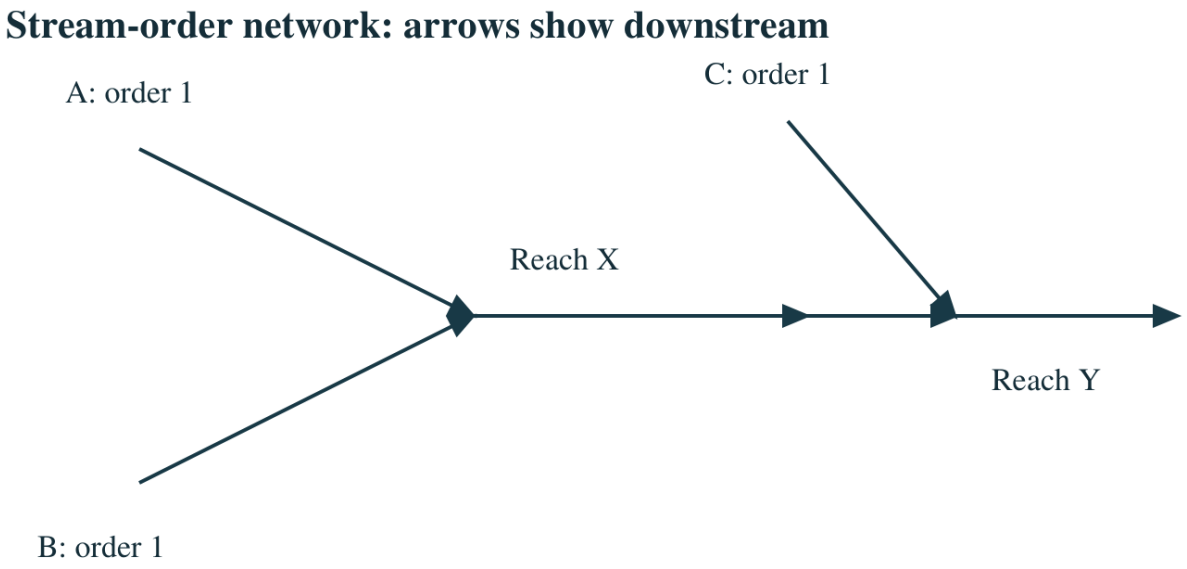
Answer: _____
_____
_____

29. (2 points) Use Figure F3. Why cannot the diagram alone determine groundwater discharge?

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Give the Strahler order of reach X.



F4. Tributary network (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Give the Strahler order of reach Y.

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. If C were second order instead, how would Y change?

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____